FIBRE PRODUCTS IN GOOD DIAGRAMS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Descent of isomorphisms. Limits, isomorphism descent lemmas of the chapter

Source reference. Original reference 0CNM: sources/stacks-chow/tags.txt, source lines 15897–15897. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 0CNP: sources/stacks-chow/tags.txt, source lines 15899–15899. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CNR: sources/stacks-chow/tags.txt, source lines 15901–15901. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0CNS: sources/stacks-chow/tags.txt, source lines 15902–15902. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CNT: sources/stacks-chow/tags.txt, source lines 15903–15903. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081E: sources/stacks-chow/tags.txt, source lines 10214–10214. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0CNM suppose that we have a cartesian diagram*

$$\begin{array}{ccc} X^1 & \xrightarrow{p} & X^3 \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle a} \\ X^2 & \xrightarrow{b} & X^4 \end{array}$$

of schemes quasi-separated and of finite type over S. For each $j = 1, 2, 3, 4$ choose $i_j \in I$ and a diagram

$$\begin{array}{ccc} X^j & \longrightarrow & W^j \\ \downarrow & & \downarrow \\ S & \longrightarrow & S_{i_j} \end{array}$$

as in (0CNP). Let $X^j = \lim_{i \geq i_j} X^j_i$ be the corresponding limit descriptions as in Lemma 0CNR. Let $(a_i)_{i \geq i_5}$, $(b_i)_{i \geq i_6}$, $(p_i)_{i \geq i_7}$, and $(q_i)_{i \geq i_8}$ be the corresponding

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morphisms of systems constructed in Lemma 0CNR. Then there exists an $i_9 \geq \max(i_5, i_6, i_7, i_8)$ *such that for* $i \geq i_9$ *we have* $a_i \circ p_i = b_i \circ q_i$ *and such that*

$$(q_i, p_i) : X_i^1 \longrightarrow X_i^2 \times_{b_i, X_i^4, a_i} X_i^3$$

is a closed immersion. If a *and* b *are flat and of finite presentation, then there exists an* $i_{10} \geq \max(i_5, i_6, i_7, i_8, i_9)$ *such that for* $i \geq i_{10}$ *the last displayed morphism is an isomorphism.*

Proof. According to the discussion in Remark 0CNS the choice of W^1 fitting into a diagram as in (0CNP) is immaterial for the truth of the lemma. Thus we may choose $W^1 = W^2 \times_{W^4} W^3$. Then it is immediate from the construction of X_i^1 that $a_i \circ p_i = b_i \circ q_i$ and that

$$(q_i, p_i) : X_i^1 \longrightarrow X_i^2 \times_{b_i, X_i^4, a_i} X_i^3$$

is a closed immersion.

If a and b are flat and of finite presentation, then so are p and q as base changes of a and b. Thus we can apply Lemma 0CNT to each of a, b, p, q, and $a \circ p = b \circ q$. It follows that there exists an $i_9 \in I$ such that

$$(q_i, p_i) : X_i^1 \to X_i^2 \times_{X_i^4} X_i^3$$

is the base change of (q_{i_9}, p_{i_9}) by the morphism by the morphism $X_i^4 \to X_{i_9}^4$ for all $i \geq i_9$. We conclude that (q_i, p_i) is an isomorphism for all sufficiently large i by Lemma 081E. □